Seat No: ______________ **Enrollment No: ____________________**

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech. Winter 2023 - 24 Examination

Semester: 3 **Date: 27/10/2023**
Subject Code: 303105210 **Time: 2:00pm to 4:30pm**
Subject Name: Computer Organization and Microprocessor **Total Marks: 60**

Instructions:
1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1	**Objective Type Questions -** (Fill in the blanks, one word answer, MCQ-not more than Five in case of MCQ) (All are compulsory) (Each of one mark)	**(15)**	**CO**	**PO**	**Bloom's Taxonomy**
	1.Write any one three byte instruction of 8085		**4**	**2**	**remenber**
	2. 8085 is _________ bit microprocessor		**1**	**1**	**remember**
	3.Can we interface 64 MB memory with 8085?(Y/N)		**5**	**4**	**Apply**
	4. Write any one maskable interrupt of 8085		**1**	**3**	**remember**
	5.________ pin is indicates that another device is requesting the use of the address and data bus of 8085		**1**	**2**	**Apply**
	6. Write the function of PC in 8085.		**1**	**1**	**Remember**
	7._____________ T state required to execute the IN 05 H instruction in 8085 a) 4 b) 7 c) 10 d) 13		**3**	**1**	**Understand**
	8. What is stack?		**2**	**2**	**remember**
	9. ALE = 1 is indicate _______________		**1**	**4**	**Undrestand**
	10. What is the high speed memory between the main memory and the CPU called? a) Register Memory b) Cache Memory c) Storage Memory d) Virtual Memory		**5**	**5**	**analyze**
	11. Flash Memory is nonvolatile memory? (Y/N) Justify your answer		**5**	**1**	**evaluate**
	12. In 8085, Which of the following instruction use to check interrupts are pending or not. a) RIM b) SIM c) LDA d) STA		**2**	**2**	**analyze**
	13. When the CY flag is set to 1 in 8085?		**1**	**1**	**Apply**
	14. Define opcode.		**4**	**2**	**Remember**
	15. Which number system is supported to 8085 processor? a) Decimal b) Hexadecimal c)Binary d) Octal		**1**	**2**	**Remember**

Q.2	**Answer the following questions**. (Attempt any three)	**(15)**			
	A) Draw and explain the programming model of 8085.		**1**	**2**	**Remember**
	B)Explain CMP M , RAL and CMA instruction of 8085		**4**	**4**	**Understand**
	C) Generate 0.5second delay using 16-bit register pair as counter method. Consider 6MHz clock frequency and the internal clock frequency is 3MHz.		**2**	**5**	**Create**
	D)Explain logical and circular shift microoperation		**4**	**3**	**Understand**
Q.3	A)Draw and explain interrupt structure of 8085	**(07)**	**1**	**2**	**remember**
	B) Draw and explain the Interfacing of 4KB EPROM to 8085 with starting address A000H.write the starting and ending location of 8085 which interface to EPROM.	**(08)**	**5**	**4**	**Create , Analyze**
	OR				
	B) Define the following (1) Machine Language (2)Assembly Language (3) Assembler (4) subroutines	**(08)**	**2**	**5**	**Remember**
Q.4	A) Write 8085 Assembly language program to convert binary to ASCII values.	**(07)**	**4**	**5**	**Create**
	OR				
	A) Explain the fetch stage; decode stage and execution stage of instruction cycle with example of any one instruction.	**(07)**	**4**	**2**	**analyze**
	B) Draw and explain Architecture of 8085 Microprocessor.	**(08)**	**1**	**1**	**Remember**